Neddy on your watch board

Part 2: put Neddy on the board and pair it with your account. Same rules as Part 1: every step says what you should see, and if anything looks different, stop and send me a photo or screenshot. **Do Part 1 first** (the board must already pass the screen test).

What you will do. (1) Get the Neddy software folder from me. (2) Install one more free add-on. (3) Type your Wi-Fi name and password into one file. (4) Press Upload. (5) Type the six-digit code the board shows into the Neddy app. (6) Hold a button and talk.

Be honest with yourself about what this is: this is the first version (0.1). I wrote it from Waveshare's own working examples, but I have never been able to run it on your actual board, so expect a bump or two. That is normal. Whatever happens, the answer is always: take a photo or screenshot and send it to me.

A. Get the Neddy software onto your PC

1. I sent you a file called `neddy_device.zip` in our chat. Click it and save it to your **Downloads** folder.

2. Open **Downloads**, right-click `neddy_device.zip` and choose **Extract All...** then **Extract**.

   > You should see: a folder named `neddy_device` containing files such as `neddy_device.ino`, `settings.h` and `es8311.c`.

3. Open File Explorer › **Documents** › **Arduino**. Move (or copy) the whole `neddy_device` folder into it, so you have `Documents\Arduino\neddy_device`.

   > The folder must keep its name `neddy_device`, and the file inside must stay `neddy_device.ino`. Arduino needs those to match.

B. Add one more free add-on (a library)

4. Open **Arduino IDE**. On the left edge click the **Library Manager** icon (it looks like a stack of books; it is the third or fourth icon).

5 In its search box type `WebSockets`. Find the one named **WebSockets** *by Markus Sattler* and click **Install**. (If Arduino asks about installing other things it depends on, click **Install all**.)

> You should see: “INSTALLED” next to WebSockets by Markus Sattler.

> Several libraries have similar names. Choose the one by **Markus Sattler**.

C. Open Neddy and type in your Wi-Fi

6 In Arduino IDE choose **File › Open...**, go to `Documents\Arduino\neddy_device` and open `neddy_device.ino`.

> You should see: a window with several tabs across the top: `neddy_device.ino`, `es8311.c`, `es8311.h`, `es8311_reg.h`, `settings.h`.

7 Click the **settings.h** tab. Near the top you will see two lines marked `<<< EDIT`:

```
#define WIFI_NAME "your Wi-Fi name"
#define WIFI_PASSWORD "your Wi-Fi password"
```

Replace only the words between the quotation marks. For example, if your Wi-Fi is called Hilton Home and the password is blue42sky:

```
#define WIFI_NAME "Hilton Home"
#define WIFI_PASSWORD "blue42sky"
```

> - Keep the quotation marks and keep everything else on the line exactly as it is.
> - Capitals and spaces matter: copy the name exactly as it appears on your phone’s Wi-Fi list.
> - The board uses **2.4 GHz** Wi-Fi only. Most home routers offer it (often the one without “5G” in its name).
> - It cannot use Wi-Fi that asks you to sign in on a web page (hotels, airports, some offices). Use your home Wi-Fi, or your phone’s hotspot.

Press `Ctrl` + `S` to save.

D. Put it on the board

8. Plug the board into the PC with the USB-C cable.

   Go to the **Tools** menu and check these exactly as in Part 1:

Tools menu item	Choose
Board	**ESP32 Arduino › ESP32S3 Dev Module**
Port	the COM port for your board (Part 1, step 15)
USB CDC On Boot	**Enabled**
Flash Size	**32MB**
PSRAM	**OPI PSRAM**
Partition Scheme	**Huge APP** (the one with the biggest app space)

9. Click the **Upload** button (round arrow pointing right, top left). Wait. The first time it can take **5 to 10 minutes** while it builds.

   You should see: the black panel at the bottom scrolling, then **Done uploading**.

   Red error text instead? Do not worry and do not change anything. Click inside the black panel, press `Ctrl` + `A` then `Ctrl` + `C`, and paste it into our chat (or send a screenshot). That is exactly what I need to fix it. If Upload cannot find the board, do the BOOT-button download-mode step from Part 1, step 18.

E. Pair the board with your Neddy account

10. When the upload finishes the board restarts by itself.

    You should see on the board's screen, in order: a green **Neddy** and "Starting", then "Joining Wi-Fi", then a black screen with **Pair me with Neddy** and a big six-digit code like **482 913**.

    Stuck on "Can't join Wi-Fi"? Check the name and password in `settings.h` (step 7), press Upload again. **"Can't reach Neddy"?** The board has Wi-Fi but cannot reach the Neddy server: check your Wi-Fi really has internet, and tell me.

11. On your Android phone, open the **Neddy app** (the same address you always use) and sign in.

12 Tap **You** (bottom right), then **Account**, then scroll down to the card called **Your devices**.

> You should see a card with a box labelled **Code on the device** and a button **Pair device**. (You see this card because you are the admin.)

Easier by voice: in a Neddy chat, say or type: *"Pair my device. The code is 4 8 2 9 1 3."* using the digits on the board.

13 Type the six digits from the board into **Code on the device**. You may name it (for example "Desk"). Tap **Pair device**.

> You should see on the phone: "Paired as … The device should let you know it's ready in a little while." On the board: **Paired!**, then "Connecting to Neddy", then a green circle with its name and **Hold BOOT to talk**.

> The code only lasts **10 minutes**. If it runs out the board shows a fresh one: use the new number.

F. Talk to it

14 Find the button labelled **BOOT** on the side of the board (the other one, **PWR**, is the power button; do not use that one). Press and **hold BOOT**, say something like *"What can you do?"*, then **let go**.

> You should see: a **red circle** and "Listening" with a small sound-level bar that moves while you talk; then an **amber** "Thinking"; then Neddy's words appear on screen and you hear her voice. At the bottom of the ready screen it shows how long she took to answer, for example "last answer 1.4 s after release".

> - Hold for at least a second. A very quick tap is ignored on purpose.
> - If the bar does not move while you talk, the microphone is not hearing you: tell me.
> - Talking while she is speaking cuts her off, like talking over a person.
> - One press can talk for up to 30 seconds.

If something goes wrong

What you see	What to do
Screen stays black after Upload	Unplug and replug. Check Tools: PSRAM **OPI PSRAM**, USB CDC On Boot **Enabled**. If still black, send me a photo and the Serial Monitor text (below).
"Can't join Wi-Fi"	Step 7 again: exact name and password, 2.4 GHz, no sign-in page. Upload again.

Code shown but the app says it isn't right	Type the six digits only (spaces are fine). If it has been over 10 minutes, use the new code the board shows.
You don't see "Your devices" in the app	Use the chat instead: say "pair my device" and read the digits.
Paired, but never gets to the green screen	Wait a minute. If it says "Neddy won't let me in", hold BOOT while plugging in the USB cable, keep holding about two seconds until it says "Pairing cleared", then pair again.
No sound from the speaker	Is a speaker attached to the board? Did Part 1's sound test (step 20) work? Tell me.
She answers with an error in words	Read it to me; for example "Talking isn't switched on yet" means something on the Neddy server needs turning on.
Want to start over	Unplug, hold BOOT while plugging in, keep holding two seconds: "Pairing cleared". Then pair again. You can also remove the device under Your devices in the app.

Sending me what the board says

★ With the board plugged in: Arduino IDE › **Tools › Serial Monitor**. Set the speed box (bottom right) to **115200**. Unplug and replug the board, then use it. Lines like `[wifi] joined`, `[pair] show code`, `[neddy] connected`, `[talk] start` and `[latency] first sound 1380 ms after release` appear. Select all, copy, and paste it into our chat. That text tells me exactly what the board did.

Good to know. The connection to Neddy is encrypted, but in this early version the board does not yet check the server's identity (I will add that before anyone else uses it). The board remembers its pairing, so after the first time you only plug it in. Your Wi-Fi password is stored in the software on the board, so keep that board with people you trust, and remove it from **Your devices** if you lose it.